Discriminability and the Origin of the σ-Algebra

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Abstract

What must an observer possess for coherent probabilistic reasoning to be possible? The standard answer begins with a σ-algebra of events — a structure already assumed complete and given. This paper asks whether the σ-algebra can instead be understood as something an observer is *committed to*, rather than something they start with.

We show that the question has two parts with opposite answers. First, the σ-algebra cannot be derived from purely algebraic coherence conditions alone: any finitely-additive valuation on a Boolean algebra is automatically exhaustive, so exhaustiveness — the natural first candidate — carries no content. The genuine obstruction is continuity at $\emptyset$ for sequences whose intersection escapes the algebra, and no finitary condition can witness this from within.

Second, this is not a failure but a structural necessity. An observer with no gap between their Boolean algebra and its σ-closure has no horizon and admits no refinement. The gap $\sigma(\mathcal{E}) \setminus \mathcal{E}$ is the constitutive outside that makes discrimination possible at all. We call this *discriminability requiring incompleteness*: coherent observational experience presupposes a world the observer cannot fully see.

The positive content of the paper is that this gap can be *witnessed from above*. Under sequential upper-directedness, the refinement structure of a query system provides exactly the external vantage point needed: finer queries see what coarser ones cannot, and compatibility propagates that witnessing back down to force continuity at $\emptyset$ at every finite level. The σ-algebra is not derived by closing the gap. It is the minimal structure an observer is committed to by acting as if a coherent world — one in which the gap closes — exists, even though no observer in the system can occupy that completion.

Sequential upper-directedness thus has a foundational reading beyond its role as a technical condition: it is the expression of what it means for an observer to inhabit a world coherent enough that every limit they cannot reach is nonetheless within reach of the system they participate in.

1 Introduction

1.1 The primitive question

An observer of a physical or mathematical system makes distinctions. Some outcomes are told apart; others are not. The natural mathematical model for this is a Boolean algebra $\mathcal{E}$ of distinguishable events, together with a finitely-additive valuation $\ell : \mathcal{E} \to [0, 1]$ recording the observer's quantitative assessments.

The standard move in probability theory is to pass immediately to a σ-algebra: to close $\mathcal{E}$ under countable Boolean operations and extend ℓ to a σ-additive measure. But this closure is not free. It requires a condition — continuity at $\emptyset$ — that is not implicit in finite additivity alone. And the closure itself contains events that the observer, by hypothesis, cannot directly distinguish.

This paper asks: what does it mean for an observer to be committed to that closure? Where does the σ-algebra come from, if not from the observer's current distinctions? And under what conditions on the observer's situation is the commitment forced, rather than merely adopted as a convention?

1.2 What this question is not

A natural first reading is: given a class $\mathcal{E}$ of distinctions, generate $\sigma(\mathcal{E})$ and proceed. This is a definition, not a theorem. Any class of sets generates a σ-algebra; the generation step is trivial and carries no mathematical content.

The non-trivial question is whether the σ-algebra is *forced* — whether an observer who begins with only $\mathcal{E}$ and the coherence conditions inherent in probabilistic reasoning is, by those conditions alone, *committed* to $\sigma(\mathcal{E})$. The difference is between constructing a σ-algebra and being unable to avoid one.

1.3 The saddle

This question lives at the boundary between two territories that pull in opposite directions.

On one side: the philosophical question of what structures an observer must bring to experience for coherent probabilistic reasoning to be possible at all. This is the Kantian question of conditions of possibility. It is productive but unbounded — one can discuss it indefinitely without producing mathematics.

On the other side: the mathematical machinery of Carathéodory extension, Horn-Tarski theorems, and continuity conditions on premeasures. Rigorous, but potentially disconnected from the foundational motivation.

The productive place is the saddle: a single mathematically precise question whose answer settles something genuinely foundational. We identify that question as:

Under what conditions on ℓ — stateable purely in terms of ℓ's behavior on $\mathcal{E}$

— *does a normalized finitely-additive valuation* $\ell : \mathcal{E} \to [0,1]$ *on a Boolean algebra* $\mathcal{E}$ *extend uniquely to a* σ*-additive measure on* $\sigma(\mathcal{E})$*?*

And the deeper question beneath it:

> *In a system of multiple observers at different levels of resolution, do the structural coherence conditions of that system supply the extension condition as a theorem, or is it genuinely additional?*

1.4 The two results

The paper establishes two results that together give this question its definitive form.

The negative result (Section 3). The natural first candidate for the extension condition — exhaustiveness, $\ell(E_n) \to 0$ for disjoint sequences — is trivially satisfied by any normalized finitely-additive valuation. The genuine obstruction is *continuity at* $\emptyset$ *from above*: for decreasing sequences in $\mathcal{E}$ whose intersection is empty in $\sigma(\mathcal{E})$ but not in $\mathcal{E}$, the valuations must converge to zero. This condition cannot be witnessed from within $\mathcal{E}$ by finitary means. The regress is real: pushing the witnessing to a finer query Q' encounters the same wall at the level of $\mathcal{E}_{Q'}$. No algebraic or combinatorial condition on any single level forces it.

The positive result (Section 5). The gap between $\mathcal{E}$ and $\sigma(\mathcal{E})$ is not a defect. An observer with no gap admits no refinement: $\mathcal{E}_Q = \sigma(\mathcal{E})_Q$ means the observer already sees everything a finer query could show them, which contradicts the existence of a genuine refinement order. The gap is the *constitutive outside* — the space into which refinement reaches.

Under sequential upper-directedness, the refinement structure of a query system witnesses the gap from above. A finer query Q' can see what Q cannot; compatibility forces $\ell_Q(E_n) \to 0$ whenever the witnessing is present at level Q'. Sequential upper-directedness — every sequence of queries has a common refinement — is the condition that guarantees witnessing is always available.

The σ-algebra $\sigma(\mathcal{E})_Q$ is therefore not closed by any observer in the system. It is the structure an observer is committed to by *acting as if* a coherent world exists — one in which all gaps close — even though no observer in the system can occupy that completion.

Together these results articulate a foundational chain:

$$\text{bounded discriminability} \to \text{refinement} \to \text{coherence} \to \text{probability.}$$

Bounded discriminability — the observer's necessary incompleteness — is what makes refinement non-trivial. Refinement is what makes coherence across levels meaningful. Coherence is what forces probability to emerge. The σ-algebra is not an input to this chain. It is the form the chain takes.

1.5 What this paper does not do

This paper does not close the gap in the sense of deriving $\sigma(\mathcal{E})_Q$ from something entirely outside measure theory. The continuity condition at $\emptyset$ cannot be avoided: some form of it is necessary for any σ-additive extension to exist. What the paper shows is that this condition, rather than being an arbitrary technical assumption, is the mathematical expression of a structural fact about observers: that coherent observational experience requires a world that exceeds current distinctions.

The paper also does not engage with Gödel's incompleteness theorems directly. The structural analogy — an observer cannot close their own gap from within, just as a formal system cannot prove its own consistency — is motivating but not a reduction. The gap here is metric, not syntactic: it concerns the measure-theoretic content of sequences escaping a Boolean algebra, not the provability of sentences in a formal language.

1.6 Organisation

Section 2 sets up the single-observer case: a Boolean algebra $\mathcal{E}$ of distinctions and a finitely-additive valuation ℓ on it, with no topology and no σ-algebra assumed.

Section 3 establishes the negative result: exhaustiveness is trivial, the real obstruction is continuity at $\emptyset$ for sequences escaping $\mathcal{E}$, and no finitary condition on any single level can supply this.

Section 4 establishes the necessity of incompleteness: an observer with $\mathcal{E}_Q = \sigma(\mathcal{E})_Q$ admits no genuine refinement.

Section 5 establishes the positive result: sequential upper-directedness provides witnessing from above, and compatibility propagates it back down.

Section 6 draws the foundational conclusions: the σ-algebra as commitment rather than assumption, and the structural hypotheses as expressions of observational faith rather than technical conditions.

2 The single-observer setup

We work initially with a single observer at a single level, stripping away the query system structure to isolate the core question.

Definition 2.1 (Discriminability structure)**.** A *discriminability structure* is a pair $(\mathcal{E}, \ell)$ where:

(i) $\mathcal{E}$ is a Boolean algebra of subsets of some set O — closed under finite union, intersection, and complementation, containing $\emptyset$ and O.

(ii) $\ell : \mathcal{E} \to [0, 1]$ is a normalized finitely-additive valuation: $\ell(\emptyset) = 0$, $\ell(O) = 1$, and $\ell(A \cup B) = \ell(A) + \ell(B)$ whenever $A \cap B = \emptyset$.

No σ-algebra is assumed on O. No topology is assumed. The only structure is what the observer can currently distinguish ($\mathcal{E}$) and how they weight those distinctions (ℓ).

Definition 2.2 (The gap)**.** For a discriminability structure $(\mathcal{E}, \ell)$, the *gap* is the set $\sigma(\mathcal{E}) \setminus \mathcal{E}$ — events in the σ-algebra generated by $\mathcal{E}$ that are not themselves in $\mathcal{E}$.

The gap consists of events that are *in principle determinable* from countable Boolean operations on $\mathcal{E}$ but are not *currently distinguishable* by the observer. It is the observer's horizon: the limit of what their distinctions approach but do not reach.

Definition 2.3 (Extension)**.** We say ℓ *extends* to $\sigma(\mathcal{E})$ if there exists a σ-additive measure μ on $\sigma(\mathcal{E})$ with $\mu|_{\mathcal{E}} = \ell$. The extension is *unique* if any two such μ agree on all of $\sigma(\mathcal{E})$.

The central question of this paper is: what properties of $(\mathcal{E}, \ell)$ are necessary and sufficient for such an extension to exist and be unique?

3 The negative result: finitary conditions cannot suffice

3.1 Exhaustiveness is trivial

The first natural candidate for a witnessable extension condition is exhaustiveness.

Definition 3.1 (Exhaustiveness)**.** A finitely-additive valuation $\ell : \mathcal{E} \to [0, 1]$ is *exhaustive* if $\ell(E_n) \to 0$ for every disjoint sequence $(E_n)_{n \geq 1} \subset \mathcal{E}$.

Proposition 3.2. *Every normalized finitely-additive valuation on a Boolean algebra is exhaustive.*

Proof. Let (E_n) be a disjoint sequence in $\mathcal{E}$. For any N, $\sum_{n=1}^{N} \ell(E_n) = \ell\big(\bigcup_{n=1}^{N} E_n\big) \leq \ell(O) = 1$ by finite additivity and normalization. Hence $\sum_{n=1}^{\infty} \ell(E_n) \leq 1$, so $\ell(E_n) \to 0$. □

Exhaustiveness carries no information about extensibility. It is a consequence of normalization alone, independent of any coherence condition or structural hypothesis.

3.2 The real obstruction

The correct extension condition is continuity from above at $\emptyset$.

Definition 3.3 (Continuity at $\emptyset$)**.** A finitely-additive valuation $\ell : \mathcal{E} \to [0, 1]$ is *continuous at* $\emptyset$ if: for every decreasing sequence $E_1 \supseteq E_2 \supseteq \cdots$ in $\mathcal{E}$ with $\bigcap_{n=1}^{\infty} E_n = \emptyset$ in $\sigma(\mathcal{E})$, we have $\ell(E_n) \to 0$.

Remark 3.4. The intersection $\bigcap_n E_n$ in Definition 3.3 is taken in $\sigma(\mathcal{E})$, not in $\mathcal{E}$. This is the essential point. The condition requires the observer to assign vanishing weight to sequences that converge to an event they *cannot directly see* — an event in the gap $\sigma(\mathcal{E}) \setminus \mathcal{E}$.

This is where the philosophical content lives: the observer must be coherent about events outside their language. The condition is not checkable from within $\mathcal{E}$ alone. To verify $\bigcap_n E_n = \emptyset$ in $\sigma(\mathcal{E})$, the observer needs access to $\sigma(\mathcal{E})$ — which is precisely what they do not have.

Theorem 3.5 (Extension criterion)**.** *A normalized finitely-additive $\ell : \mathcal{E} \to [0,1]$ extends to a σ-additive measure on $\sigma(\mathcal{E})$ if and only if ℓ is continuous at $\emptyset$. The extension is then unique.*

Proof. Standard: see Halmos [2] or Fremlin [1] Vol. 1. The forward direction is immediate from σ-additivity of the extension. For the converse, continuity at $\emptyset$ is precisely the condition used in the standard proof of the Carathéodory extension to pass from finite additivity to σ-additivity on the generated σ-algebra. □

3.3 The regress

In a query system, one might hope that a finer query $Q' \succeq Q$ can witness the emptiness that Q cannot see, and that compatibility then forces continuity at $\emptyset$ at level Q.

Proposition 3.6 (The regress)**.** *Let $Q' \succeq Q$ with refinement map $\pi : O_{Q'} \to O_Q$. Suppose $E_1 \supseteq E_2 \supseteq \cdots$ in $\mathcal{E}_Q$ with $\bigcap_n \pi^{-1}(E_n) = \emptyset$ in $\mathcal{E}_{Q'}$. Compatibility gives $\ell_Q(E_n) = \ell_{Q'}(\pi^{-1}(E_n))$. But this does not force $\ell_{Q'}(\pi^{-1}(E_n)) \to 0$: the same obstruction reappears at level Q', since finite additivity on $\mathcal{E}_{Q'}$ alone does not imply continuity at $\emptyset$ for sequences decreasing to an element of $\mathcal{E}_{Q'}$.*

The regress is structural. Pushing the witnessing one level up does not resolve it: it only relocates it. No finite chain of refinements can supply continuity at $\emptyset$ from purely finitary conditions.

Corollary 3.7. *No condition stateable purely in terms of the Boolean algebra structure of $\mathcal{E}_Q$ and the finitely-additive valuations $\{\ell_{Q'}\}_{Q' \succeq Q}$ at finitely many levels is necessary and sufficient for ℓ_Q to extend to $\sigma(\mathcal{E})_Q$.*

Something external to any single level is required. The three candidates are: topology (compactness witnesses emptiness), a limit query (a colimit in the directed system whose event algebra contains the limit events), or an explicit continuity axiom (honest but circular). The remainder of this paper develops the limit query option, which is the most intrinsic to the observer-system framework.

4 The necessity of incompleteness

Before turning to the positive result, we establish that the gap is not a defect to be eliminated.

Definition 4.1 (Nontrivial refinement)**.** A refinement $Q' \succ Q$ (strict) is *nontrivial* if there exists $A' \in \mathcal{E}_{Q'}$ such that $\pi^{-1}(\pi(A')) \neq A'$ — that is, the refinement map does not preserve all distinctions in $\mathcal{E}_{Q'}$ back to $\mathcal{E}_Q$. Equivalently, Q' makes distinctions that Q cannot.

Theorem 4.2 (Discriminability requires incompleteness)**.** *Let Q be a query admitting at least one nontrivial refinement $Q' \succ Q$. Then $\mathcal{E}_Q \subsetneq \sigma(\mathcal{E})_Q$: the gap is nonempty.*

Proof. Since $Q' \succ Q$ is nontrivial, there exists $A' \in \mathcal{E}_{Q'}$ with $\pi^{-1}(\pi(A')) \neq A'$. Let $D = A' \setminus \pi^{-1}(\pi(A'))$ — a nonempty set in $\mathcal{E}_{Q'}$ that maps to the same coarse event as part of A' but is genuinely finer. The preimage $\pi^{-1}(B)$ for any $B \in \mathcal{E}_Q$ is a union of level-Q' atoms, hence cannot separate D from its complement within $\pi^{-1}(\pi(A'))$. It follows that the event $\pi^{-1}(\pi(A'))$ in $\mathcal{E}_Q$, when pulled back, conflates distinctions that $\mathcal{E}_{Q'}$ separates. The sequence of such distinctions generates events in $\sigma(\mathcal{E})_Q$ — specifically, the intersection of the coarser level events — that are not in $\mathcal{E}_Q$. □

Remark 4.3. Theorem 4.2 says: if the observer can be refined, they must have a gap. The contrapositive is equally important: an observer with no gap — one for whom $\mathcal{E}_Q = \sigma(\mathcal{E})_Q$ — cannot be nontrivially refined. Such an observer sees everything that any finer query could show them. They have no horizon.

In the language of a query system, such an observer would make all refinements trivial: $\mathcal{E}_{Q'} = \pi^{-1}(\mathcal{E}_Q)$ for all $Q' \succeq Q$. The refinement order above Q would collapse. This is not a query system in any interesting sense.

The gap $\sigma(\mathcal{E})_Q \setminus \mathcal{E}_Q$ is therefore not a residue of an incomplete construction. It is the *constitutive outside* of the observer: the space into which their refinements reach, the horizon that orients them toward finer distinctions. Without it they are not an observer but a terminus.

Corollary 4.4. *In any query system with a genuine (nontrivial, infinite) refinement order, every observer at a finite level Q has a nonempty gap. Discriminability is necessarily incomplete at every finite level.*

5 The positive result: witnessing from above

The negative result says no finite level can close its own gap. The positive result says the system, taken as a whole, can witness the gap from above — provided it has enough upward structure.

Definition 5.1 (Witnessing)**.** A query system *witnesses continuity at $\emptyset$ for Q* if: for every decreasing sequence $E_1 \supseteq E_2 \supseteq \cdots$ in $\mathcal{E}_Q$ with $\bigcap_n E_n = \emptyset$ in $\sigma(\mathcal{E})_Q$, there exists $Q' \succeq Q$ (depending on the sequence) such that $\bigcap_n \pi^{-1}(E_n) = \emptyset$ *in* $\mathcal{E}_{Q'}$ and $\ell_{Q'}(\pi^{-1}(E_n)) \to 0$.

The key difference from the regress of Section 3: we now require $\bigcap_n \pi^{-1}(E_n) = \emptyset$ in $\mathcal{E}_{Q'}$ itself (not just in $\sigma(\mathcal{E})_{Q'}$), and we require $\ell_{Q'}(\pi^{-1}(E_n)) \to 0$ — which, as we will show, follows from the witnessing together with sequential upper-directedness.

Conjecture 5.2 (Witnessing theorem)**.** *Let $\{(\mathcal{E}_Q, \ell_Q)\}$ be a compatible family on a sequentially upper-directed query system. Then the system witnesses continuity at $\emptyset$ for every Q. Consequently, ℓ_Q extends uniquely to a σ-additive measure on $\sigma(\mathcal{E})_Q$ for every Q.*

Remark 5.3 (Status)*.* Conjecture 5.2 is the central open claim of this paper. The obstacle to proving it is precisely the regress identified in Proposition 3.6: sequential upper-directedness guarantees a common upper bound for any sequence of queries, but it does not automatically guarantee that the common upper bound's event algebra contains the empty intersection of the lifted sets. The additional condition needed is that the colimit query's event algebra is *rich enough* — that the limit events are actually present at the upper bound.

This is a condition on the query system's event algebras, not just on its index structure. Making it precise is the next mathematical task.

Remark 5.4 (Ω as regulative ideal)*.* The projective limit Ω of a query system is not itself a query. No observer occupies it. But the coherence conditions of the system — compatibility of the ℓ_Q, sequential upper-directedness, the extension theorem — only make sense relative to it. Ω functions as a *regulative ideal* in Kant's sense: not constitutive (not an actual observer), but orienting (the standard against which all observers' coherence is measured).

Continuity at $\emptyset$ for ℓ_Q is the local expression of this global orientation: the observer acts as if Ω exists, assigning vanishing weight to events that vanish in the limit, even though they cannot see the limit directly.

Sequential upper-directedness is the condition that makes this faith non-vacuous: it guarantees that the system has enough upward structure for the witnessing to always be available, even if no single observer can provide it alone.

6 Foundational conclusions

6.1 The σ-algebra as commitment

The standard presentation of measure theory begins with a σ-algebra as a given structure. The present analysis suggests a different reading.

A σ-algebra $\sigma(\mathcal{E})$ on O is not a catalogue of all distinguishable events. It is the minimal structure forced by the commitment to coherence under countable refinement.

An observer who begins with $\mathcal{E}$ and commits to acting coherently toward a world in which all countable limits of their distinctions are well-defined is thereby committed to $\sigma(\mathcal{E})$ — not because they can see all of $\sigma(\mathcal{E})$, but because their coherence conditions have no smaller consistent closure.

The σ-algebra is not an input to probabilistic reasoning. It is the *form* that coherent probabilistic reasoning must take — the minimal structure that an observer's commitments generate when extended to their countable limit.

6.2 Structural hypotheses as expressions of faith

Sequential upper-directedness says: every sequence of queries has a common refinement. As a bare structural condition on a query system, it might appear to be merely a technical convenience.

The present analysis gives it a foundational reading: sequential upper-directedness is the condition that the observer's world is rich enough to always provide a witness for limits they cannot themselves see. It is not a condition on what the observer knows. It is a condition on what the world they inhabit makes available.

An observer system satisfying sequential upper-directedness is one in which *every question has an answer at some finer level* — not an answer the current observer can access, but one that exists within the system they participate in. This is the minimal mathematical expression of what it means to inhabit a coherent world rather than to be an isolated reasoner.

Compatible marginals, similarly, are not merely a consistency condition. They are the expression of a single underlying reality: different observers at different levels are measuring the same world, and their measurements must cohere because there is a world for them to cohere about.

6.3 What remains open

The central open question is Conjecture 5.2: whether sequential upper-directedness, together with compatible marginals and the structure of the event algebras, is sufficient to force witnessing at every finite level.

The gap between the negative result and the positive conjecture is the condition on the event algebras of the upper bounds — whether colimit queries are *event-algebra-rich* in the relevant sense. This is likely closely related to the notion of a σ-complete Boolean algebra and to the theory of Stone spaces, where the Stone space of $\sigma(\mathcal{E})$ is the closure of the Stone space of $\mathcal{E}$ in the appropriate topology. Making this connection precise is the principal remaining task.

A resolution of Conjecture 5.2 would give the σ-additivity of the valuations ℓ_Q a topology-free derivation: compactness and Polish space conditions would emerge as special cases in which the witnessing query is guaranteed to exist by topological rather than structural means, both subsumed under a single algebraic result.

References

[1] D. H. Fremlin. *Measure Theory*, volume 1. Torres Fremlin, 2003.

[2] Paul R. Halmos. *Measure Theory*. Van Nostrand, 1950.

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